Spatial analysis of Environmental Health Risks in Seattle: An intersectional GIS approach

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Introduction

For this paper I wanted to study physical environment risk levels in Seattle through an intersectional framework. I am interested intersectionality and its application to GIS because I think that the theory is a valiant effort to capture the various reasons for observed inequalities. Looking at where marginalized identities intersect and how that influences the socio-economic status, health, or well-being of individuals is applicable for a lot of different research. Intersectionality theory has mainly been applied within qualitative research in the humanities but has been applied to health research in recent years. A 2022 report from the government of Canada assessed the frequency of intersectional indicators in the health equity literature and found that 'time and space' is underrepresented in current research (Public Health Agency of Canada, 2022). My model was largely inspired by Eco-Intersectional Multilevel modeling as proposed in the 2021 paper *Intersectional environmental justice and population health inequalities* (Alvarez and Evans, 2021). They furthered this research with another paper in 2023 refining the method. This multi-layered approach with intersectional variables being aggregated at the census-tract level allows for interesting classifications. Their studies were on the national-scale, so I want to adapt the intersectional methodology to the urban scale to identify reoccurring intersectional identities that surface as health inequity indicators.

The next aspect is to characterize the physical environment of Seattle for environmental justice. Poor, minority dense communities are often disproportionately exposed to environmental health hazards. Especially within urban environments, low-income areas may be highly walkable or even have adequate greenspace; however, larger environmental risk factors can negatively impact the physical environment's effects on resident health. GIS has been used to support this, as the 2009 study in NYC that compared GIS and field observation data to show how neighborhood conditions differed significantly between poor and nonpoor neighborhoods, even when equally walkable (Neckerman, 2009). This is why intersectional theory is highly applicable to the physical health environment in a city. Traditional metrics may omit or obscure the influence of intersecting identities on unequal environmental health risks. Environmental justice activism can utilize intersectional GIS approaches to identify which identities are harmed the most in individual urban areas.

Methods

I included four indicators to describe environmental risk for Seattle; noise exposure, water toxicity concentration, air toxicity concentration, and PM2.5 concentrations. The 2020 noise pollution data came from Seto and Huang's 2023 paper on *the National Transportation Noise Exposure Map for the United States*. Using the QGIS field calculator, an estimated population-weighted exposure (DBA) was calculated. The EPA set an average 24-hr exposure limit of 55 A-weighted decibels (dBA) to describe the level at which noise could have adverse health effects in residential areas, so the population weighted exposure can be used with that (U.S. EPA 1974). RSEI data was obtained from the Washington Tracking Network, who had calculated the 3-year average toxicity-weighted concentrations of chemical releases into air from facility emissions and off-site incineration by census tract from 2018-2021. A PM2.5 metric was also obtained from the WTN for 2014-2017 estimates from the Washington State Department of Ecology. The final exposure metric was also obtained from WTN's census tract level aggregation of 2021 EJSCREEN estimates, based on RSEI data. All four datasets were subset to

only include Seattle census tracts via attributes joined by location with a 2010 Census tract boundary shapefile in QGIS.

My intersectionality approach included a similar four-digit stratum ID method as used in the EIM model, but adapted for Seattle and this project. Demographic data came from 2019 ACS five-year estimates. The data included is population percentage black, population percentage Asian, population percent bachelor's degree or better and median household income, all from DP02, DP03, and DP05. 2019 ACS data was chosen because it fits best within the temporal ranges of the environmental exposure data I had access to. The other data included is from the Location Affordability Index (v.3) from the HUD. This uses data from the 2012-2016 ACS which is then run through their model to separate results via different living categories. I used 3 category 1 data for household metrics; modeled housing cost, modeled transportation cost, and control household income. The racial ID consists of four categories: (1) census tracts below the median percent Black and below the median percent Asian residents; (2) census tracts above the median percent Black and below the median percent Asian residents; (3) census tracts below the median percent Black and above the median percent Asian residents (4) census tracts above the median percent Black and above the median percent Asian residents. Asian population percentage was chosen instead of Latinx because the large Asian population in Seattle and the low Latinx population made it a more viable choice for this scale of analysis. This variable was made using the field calculator in QGIS. Educational attainment is the percent of residents over the age of 25 with at least a bachelor's degree and consists of 3 terciles; (1) highest; (2) middle; (3) lowest. This variable was made using the field calculator in QGIS. Median Household Income was also distributed in terciles; (1) highest; (2) middle; (3) lowest, also done in the QGIS field calculator. The final category was made using the LAI data to categorize each census tract by affordability; (1) (modeled housing cost * 12 + modeled transportation cost) / control median household income (73.4k) >= 0.45; (0)(modeled housing cost * 12 + modeled transportation cost) / control median household income < 0.45. The sum of transportation and housing costs as a percentage of the median household income of Seattle as a whole was compared to 45% because of the common recommendation of 30% of income on housing and 15% on transportation. This metric was made as more of a spatial classification (such as rural/urban). In (1) census tracts the cost of living is likely higher than the median Seattle resident can afford, and inverse for (0) census tracts. This intersectional strata was developed using SDOH concepts from the aforementioned study and classic intersectionality literature on 'Synergies of Oppression', to include social and economic identities and geographies (McPherson, McGibbon, 2010). Finally, the data was aggregated into one .geojson file using spatial joins in QGIS. This made the data easy to use and all aggregated into census tracts. In my intersectional approach, this helped assess the relevancy of the IDs by allowing for easy comparison with the raw metrics. Since spatially extensive Environmental

Results

Choropleth maps were made of each of the strata as seen in Figure 1. One merit of this classification system is that it allowed for uniform analysis. Visualized through the maps, there seems to be distinct spatial clustering of environmental risks and socio-economic variables in Seattle. A surprising result was the significant clustering between the different environmental risk measures. Their choropleths showed consistently high exposure risk for census tracts on the southeastern side of Seattle. Noise pollution was the narrowest of the environmental risks, with the cluster of high values not extending West as does for RSEI, wastewater, and PM2.5

measures. The visual clustering pattern for noise pollution is best reflected by racial identity and educational attainment of a tract, meaning there is a chance that those specific intersectional

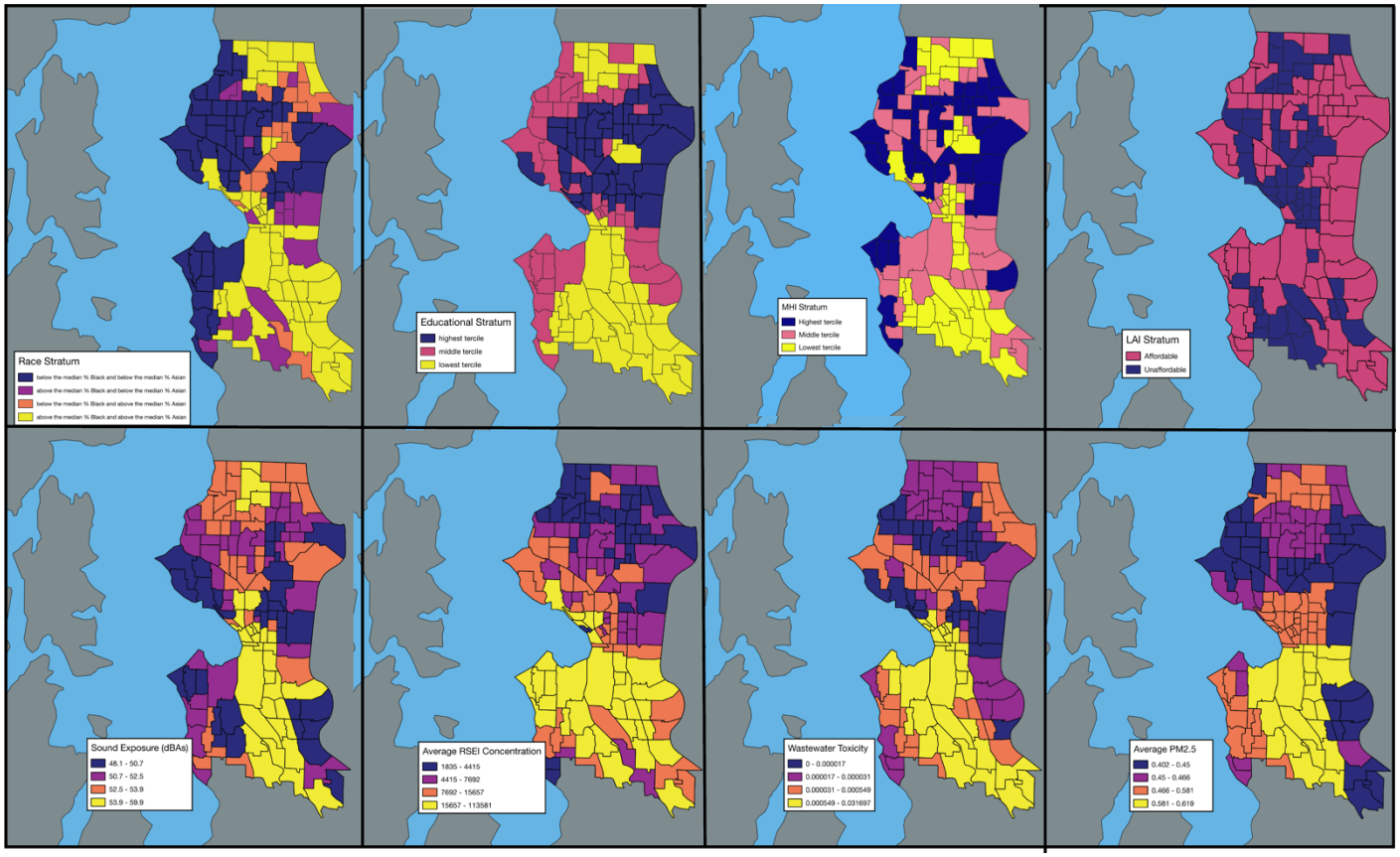

Figure 1. Social Strata

identities have an unique noise exposure risk in Seattle.

Next, descriptive statistics were calculated and a scatterplot matrix was made. From the scatterplot matrix you can assess the efficacy of each intersectional strata on each environmental risk. Noise exposure and PM2.5 concentrations both showed statistical significance across all four categories. These relationships were nearly all positive; meaning as the stratum values increase exposure risk also increases.

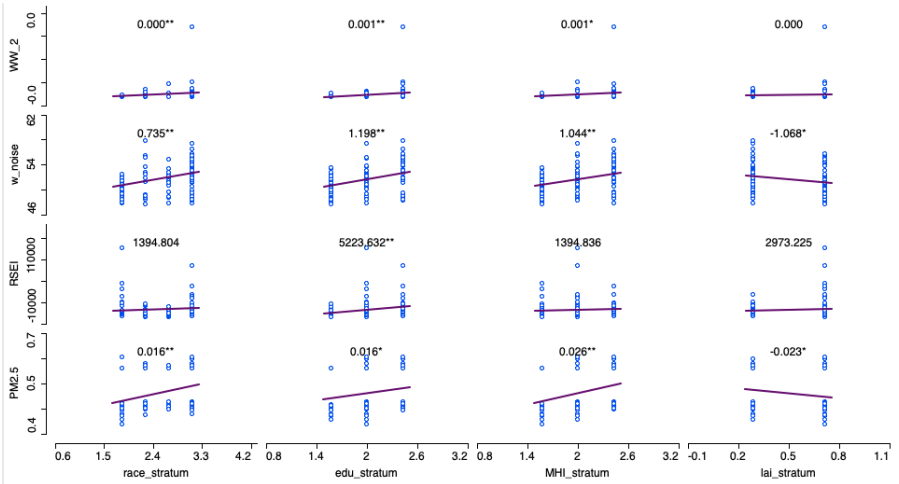

Figure 2. Scatter plot matrix of social stratum against environmental exposures

Since the stratum were designed with the greater values representing traditionally marginalized identities, this supports the hypothesis.

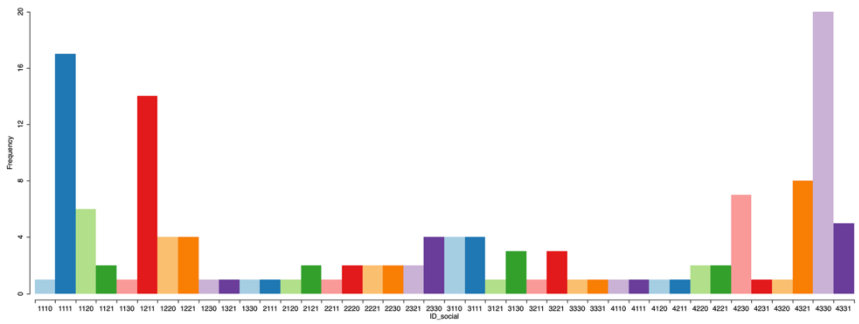

Figure 3. Census Tract Stratum ID frequency in Seattle, WA

Figure 3. shows the distribution of the compiled intersectional IDs across Seattle census tracts. From this we see that 1111 and 4330 are the most common compositions of census tracts in Seattle. This alludes to strong segregation in Seattle along racial, economic, educational, and situational identities. If a census tract shows 1 positive intersectional identity, it is likely the area will perform positively for other intersectional identities. This was investigated further with principle component analysis.

Multivariate principle component analysis was an interesting method to apply to intersectional data as it can reduce the dimensions in accordance to the data pair with it. This seemed to work well with the environmental exposure metrics. PCA clustering was done including the same 8 variables as in Figure 1.The results provided three valuable indicators of variance. PC1 seemed to capture socioeconomic indicators with greatest influence being median household income. PC2 showed high contribution from RSEI and affordability. This could reflect either the urban structure of Seattle or environmental exposure, indicating that the cost of living in Seattle could directly correlate to lower toxic exposure levels. I brushed the 3rd quartile of PC1 in a box plot, with a parallel coordinate plot (Figure 3) of the top 4 most influential variables; MHI_stratum, edu_stratum, race_stratum, and noise pollution. This revealed a distinct difference between the levels of noise pollution for census tracts with higher stratum IDs to that of lower. That is census tracts that had the lowest median household income and the lowest educational attainment rates, had significantly greater noise exposure. Not only that, the significant contribution of the race to exposure risk has been revealed. The first racial stratum—the most Caucasian—had significantly lower average distribution for noise levels. The parallel coordinate shows that census tracts with above the median population % black and Asian (the two largest minority groups in Seattle) have a greater risk for noise pollution, and explained the variance across all included variables the best.

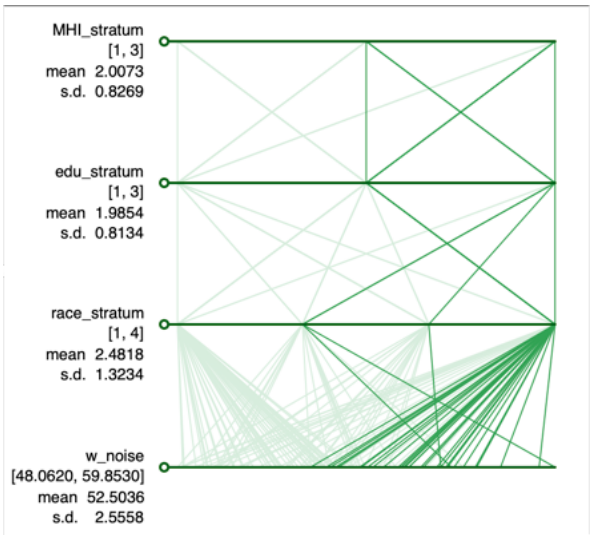

Figure 4. Parallel coordinate plot between correlated census tract identities

PC2, the environmental principle component, was similarly analyzed with RSEI concentrations, affordability, PM2.5 concentrations, and waste water toxicity levels. As indicated by their PC2 coefficients, the correlation between high RSEI concentrations and affordability explained variance with more distinct clusters. Furthermore, plotting both PC1 and PC2 on a scatter plot reveals the outlying census tracts that best explain the variance in social intersections and physical environmental exposures in Seattle. The strip of tracts extending southeast from the Pugent sound to Seattle's bottom corner can be identified as a distinct cluster apart from the rest of the city.

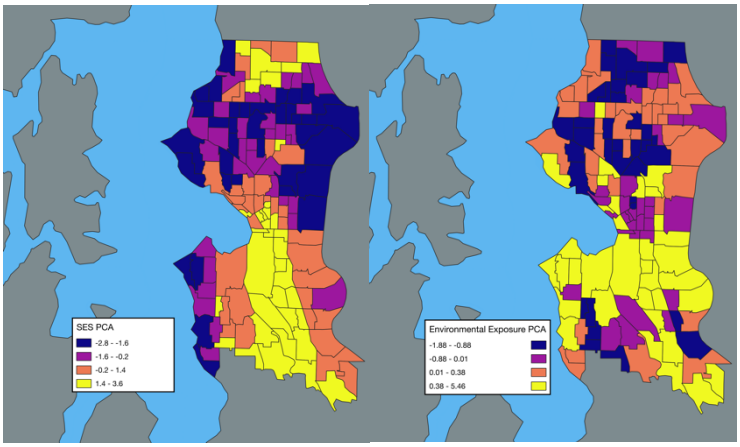

Figure 5. SES and Environmental Exposure PCA variable choropleth

Discussion

This analysis supported the efficacy of the intersectional methodology applied for physical environmental exposures in Seattle. The tracts with the highest environmental exposure risks were consistently the tracts with the most SES intersection. The distinct spatial distribution of socioeconomic and environmental risks suggests targeted urban planning and environmental policies could mitigate the compounded negative effects in lower SES areas. By focusing redevelopment efforts and environmental clean-up initiatives on these high-risk tracts, Seattle could reduce health disparities stemming from unequal environmental burdens. This data set could also aid in further research, including various health outcomes, to see if these results agree spatially and/or demographically. Stratum ID could be a quick way to assess the intersectional profile of a census tract of interest. The results of this paper show exposure to risk disparities in Seattle are reflected by the most common intersectional strata. 4330 was the most plentiful census tract ID in Seattle, and these tracts also fit the distribution of locations with greater exposure risks. Identifying the most reoccurring identities for areas of health disparity in urban areas can help us better understand and improve healthcare access and needs.

The distinct spatial clustering of environmental risks was consistent in Seattle. Although there was still some variation between the environmental measures analyzed, there is a spatial trend of census tracts that have worse air quality, noise pollution, and potentially water quality. This is not an uncommon trend in urban areas but the differences in Seattle appear to be stark and along one area. In general, the air quality risks are worse near the bay which is to be expected. However, there is a distinct difference between the severity for the census tracts on the northern portion of Elliot bay compared to the southern portion. This is where the industrial center of Seattle is and has been. This is an obvious explanation for the distribution of RSEI

toxicity, wastewater pollution, and PM2.5 concentrations. Industrial areas have historically been the areas of cities where displaced marginalized groups have had to settle. Racist, classist policies such as redlining have historically influenced urban development that further harms communities of marginalized identities with lasting health disparities.

Noise pollution can also be explained by the industrialized area of the southern bay, but further inspection reveals that the industrial ports are not the primary culprit for this negative environmental health risk. Not only is the southern portion of Seattle home to industry, it is also home to two airports. This is likely the actual main reason for increased noise pollution in these tracts. Residential areas around airports are also historically undesirable because of the negative externalities such as the noise pollution captured here. The only census tracts that have sound levels that exceed 55 dBA limit are not just the tracts closest to the airports, but also tracts that experience the most other health risks, socially and environmentally. Noise pollution can severely degrade the well-being of individuals in affected residential areas; sleep loss and other mental health effects often exacerbate other health concerns.

The goals for this analysis were met to a certain degree, however, efforts should be made to include more granular data on other social determinants of health, which could unveil more complex interactions and provide a clearer picture of the multifaceted aspects of environmental justice issues. Recognizing the limitations of the current approach, particularly the reliance on available data that may not capture all relevant variables, it's clear that a more comprehensive model could enhance the understanding and address the nuances of intersectionality in environmental exposures. Most of the time on this project was spent trying to source data with high enough granularity and during a manageable timeframe. This is a major drawback of an intersectional approach such as this; more measurements included means that there is a decreased likelihood that all of the variables can be from the same year. This limits any meaningful analysis of precise intersections because data can change over the years. The variables used in this analysis are also specifically at high risk of variation. The demographics of census tracts can sometimes change rather rapidly. The margin of errors compound as you add more strata to the analysis, making the results less accurate as data is increased.

Conclusion

Attempting to apply an intersectional approach to characterize disparities in environmental health exposures across an urban area was not a straightforward task that proved to have many limitations and flaws. Extensive data wrangling and accurate data sources are important to get accurate measures when aggregating many datasets onto one spatial unit. I believe the concept was supported by statistical and spatial analysis, but application to other cities may not be as effective. Intersectional spatial classifications could certainly aid in identifying how specific identities are at risk, in specific cities. Tailoring the intersectional components included to the area of analysis proved valuable for the efficacy of this method. Government agencies should consider this approach to identify the specific social, economic, environmental, racial, and components of census tracts that may contribute to health disparities and outcomes. A unique patient profile is essential for a doctor to diagnose ailments and then design specific health plans. Using intersectional identities for neighborhoods or census tracts can serve as a guide for planners and officials to help diagnose urban issues and hopefully design specific remedies.

Sources and References

Alvarez, C. H., & Evans, C. R. (2021). Intersectional environmental justice and population health inequalities: A novel approach. Social Science & Medicine, 269, 113559. https://doi.org/10.1016/j.socscimed.2020.113559

Alvarez, C. H., Calasanti, A., Evans, C. R., & Ard, K. (2022). Intersectional inequalities in industrial air toxics exposure in the United States. Health & Place, 77, 102886. https://doi.org/10.1016/j.healthplace.2022.102886

McGibbon, E., & McPherson, C. (n.d.). Applying Intersectionality & Complexity Theory to Address the Social Determinants of Women's Health.

Neckerman, K. M., Lovasi, G. S., Davies, S., Purciel, M., Quinn, J., Feder, E., Raghunath, N., Wasserman, B., & Rundle, A. (2009). Disparities in Urban Neighborhood Conditions: Evidence from GIS Measures and Field Observation in New York City. Journal of Public Health Policy, 30(1), S264–S285. https://doi.org/10.1057/jphp.2008.47

Public Health Agency of Canada. (2022, June 13). How to integrate intersectionality theory in quantitative health equity analysis? A rapid review and checklist of promising practices. https://www.canada.ca/en/public-health/services/publications/science-research-data/how-integrate-intersectionality-theory-quantitative-health-equity-analysis.html#fn92

Seto, K., & Huang, L. (2023). National Transportation Noise Exposure Map for the United States. Retrieved from https://www.medrxiv.org/content/10.1101/2023.02.02.23285396v1

Washington Tracking Network. (2021). Risk-Screening Environmental Indicators (RSEI) data for chemical releases 2018-2021. US EPA EJSCREEN 2021, based on RSEI 2018 - 2020 3-year average. Retrieved from https://fortress.wa.gov/doh/wtn/

Washington State Department of Ecology. (2017). PM2.5 concentration estimates 2014-2017. Retrieved from https://fortress.wa.gov/doh/wtn/

Washington Tracking Network. (2021). EJSCREEN, based on RSEI: Environmental Justice Screening and Mapping Tool, 2021 Data. Retrieved from https://fortress.wa.gov/doh/wtn/

U.S. Census Bureau. (2019). American Community Survey (ACS) five-year estimates, DP02, DP03, and DP05.

U.S. Department of Housing and Urban Development (HUD). (2016). Location Affordability Index, Version 3, based on 2012-2016 ACS data. [USCB]

U.S. Environmental Protection Agency (EPA). (1974). Noise pollution and abatement guidelines.